Diploma in

Data Analytics

Lesson 1: Introduction to R

- Introduction to R
- Working in R
- Maximum likelihood

Lesson Objectives

Lesson 1

Introduction to R

ONE

What is R

Powerful statistical programming language and free software for data analysis

Why R

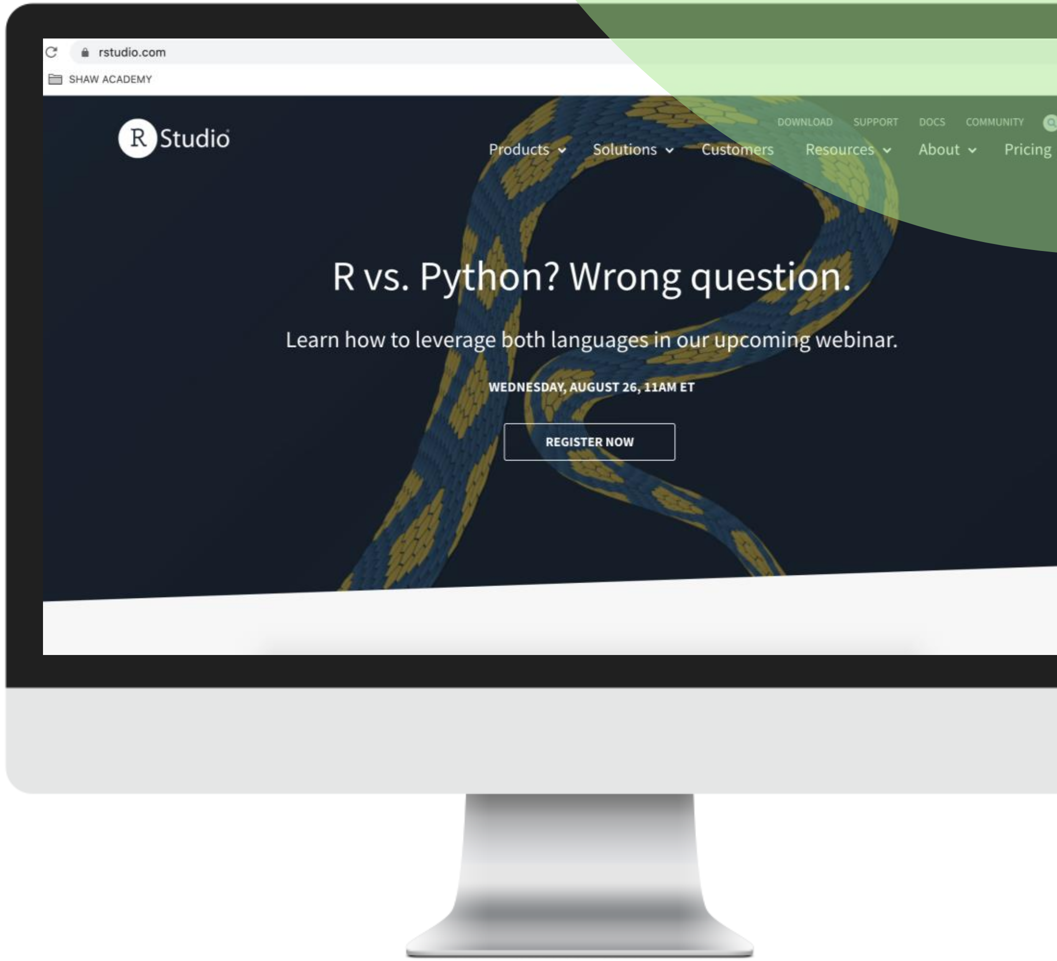
- Open source
- Ranks 8th in TIOBE index (and increasing)
- Windows, Mac, Linux OS
- Libraries implements wide variety of statistical and graphical techniques

Download R

http://cran.r-project.org/

R Studio

IDE or *Front face* for R



Why RStudio

- Open source
- Windows, Mac, Linux OS
- User-friendly

Download RStudio

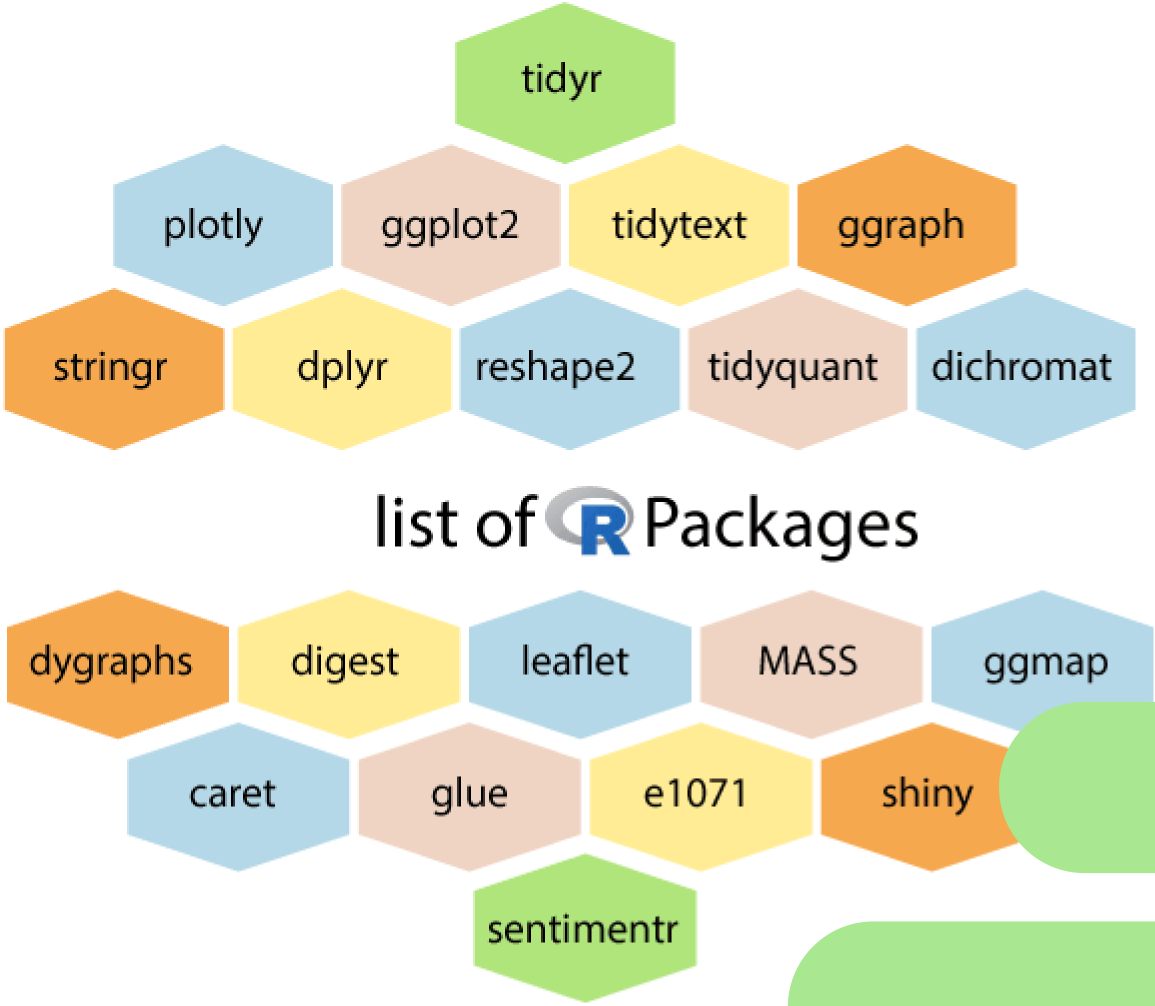
https://rstudio.com/products/rstudio/download/

Working in R

TWO

R packages

- Over 15 000 packages
- Most used and popular packages: tidyr, plotly, ggplot2, etc.



Source: https://www.javatpoint.com/list-of-r-packages

ISLR library

Package for collection of
dataset used in book ISLR

Maximum likelihood

THREE

Estimation

- Point
- Interval



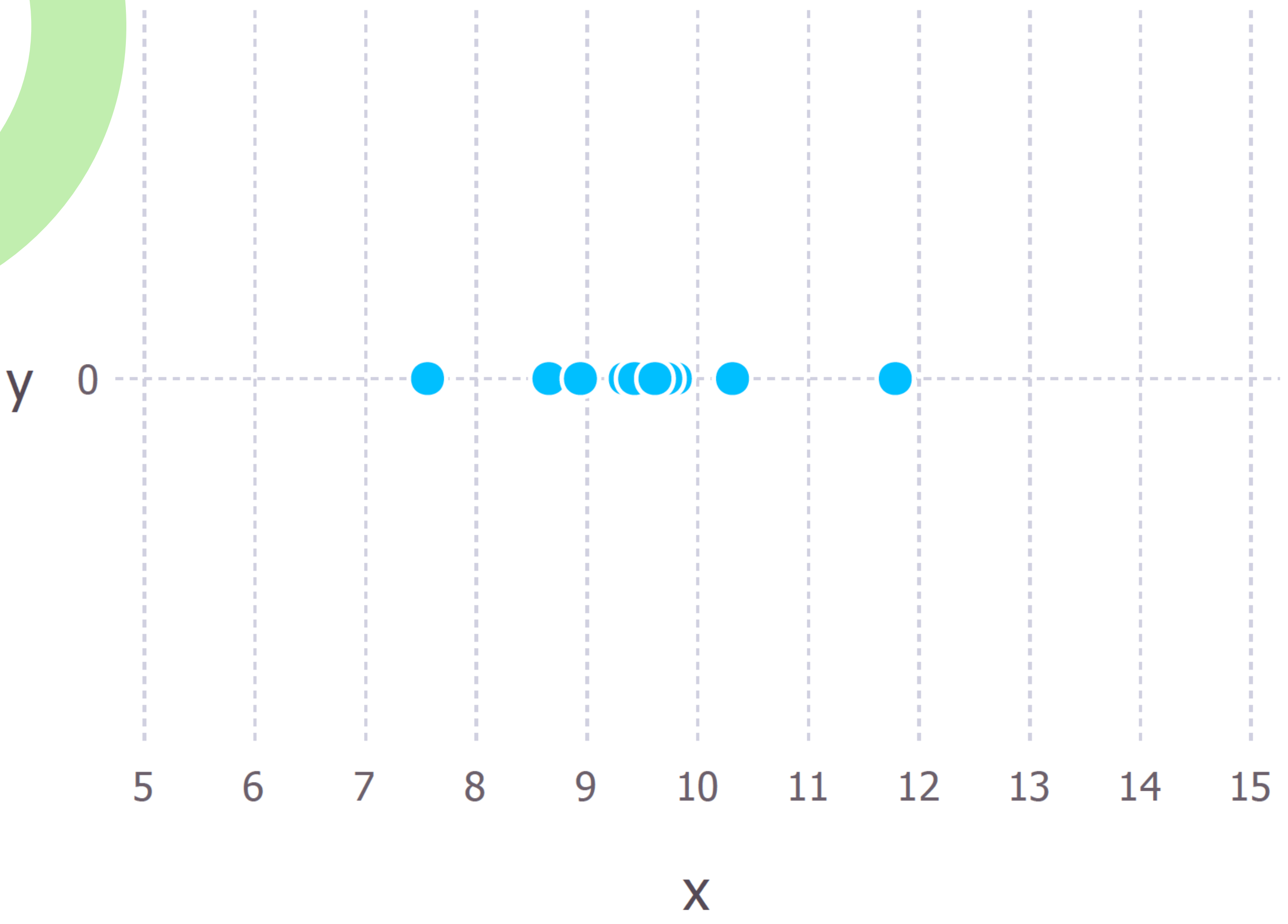
Methods of estimation

Method of least squares

Maximum likelihood

Maximum likelihood

Method used to estimate the parameters of a distribution



Maximum likelihood estimation

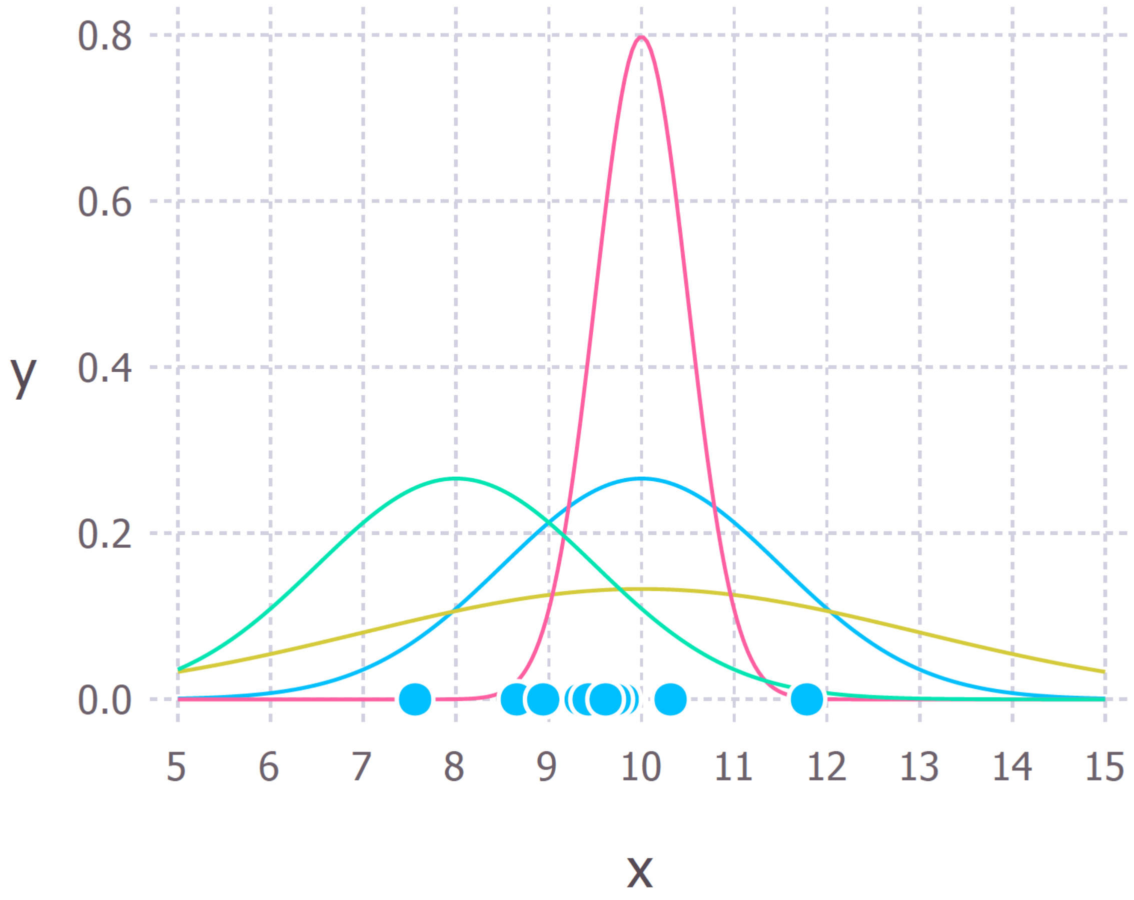
Which model fits the data the best?

Source: https://towardsdatascience.com/probability-concepts-explained-maximum-likelihood-estimation-c7b4342fdbb1

Maximum likelihood estimation

Normal distribution

Blue line is best fit for data points



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